H2

ANDERSON JUNIOR COLLEGE
HIGHER 2

BIOLOGY 9648/01

Paper 1 Multiple Choice

21 September 2015
Monday

1 hour 15 min

Additional Materials: Multiple Choice Answer Paper

READ THESE INSTRUCTIONS FIRST

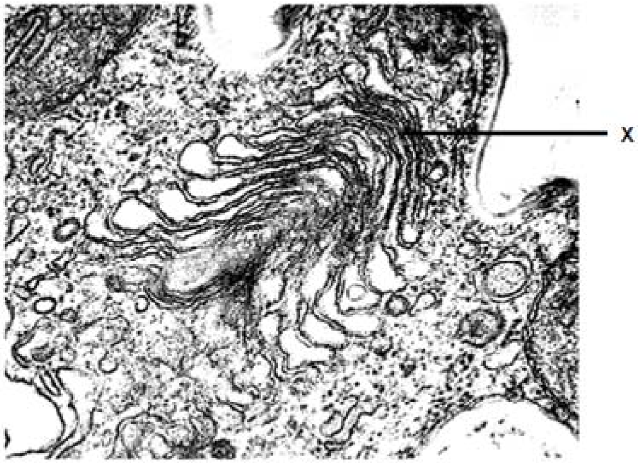
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, PDG and identification number on the Answer Sheet.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Calculators may be used.

1 Which of the following are the most likely consequences for a cell lacking structure **X**?



1 The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.

2 The cell dies from a lack of enzymes to digest food taken in by endocytosis.

3 The cell dies from the accumulation of worn out organelles within itself.

4 The cell is unable to reproduce itself.

5 The cell is unable to export its enzymes or peptide hormones.

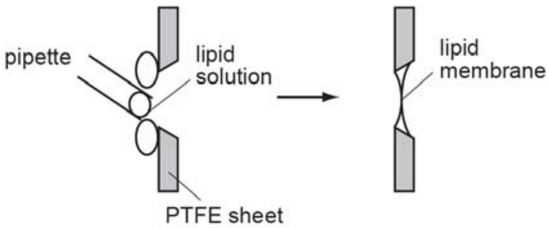
A 2 and 3

B 2 and 5

C 3, 4 and 5

D 1, 2, 3 and 5

2 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.



Such a lipid membrane is impermeable to water soluble materials including charged ions such as Na^+ or K^+.

In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added to the membrane, at which time current flowed.

What kind of molecule is gramicidin?

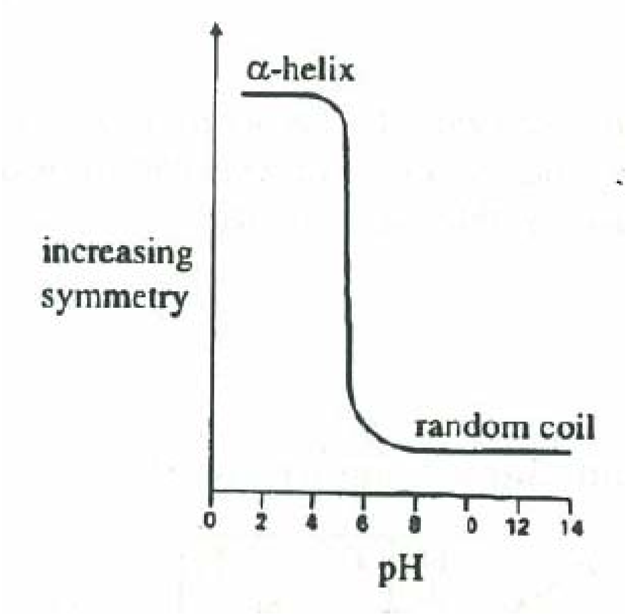
A A carbohydrate molecule found only on the outside of the membrane.

B A non-polar lipid which passes all the way through the membrane.

C A protein molecule with both hydrophilic and hydrophobic regions.

D A protein molecule which has only hydrophobic regions.

3 The graph shows the effect of pH on the structure of a protein which consists entirely of repeating residues of one amino acid.



Which of the following statements is true?

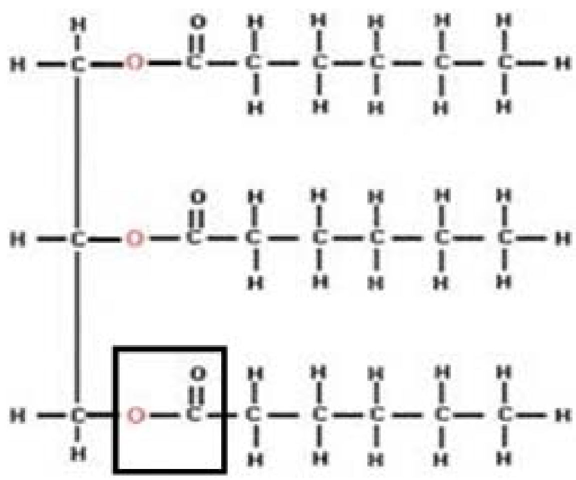
A At high acidity, the protein loses its primary structure.

B At high acidity, the protein loses its secondary structure.

C At low acidity, the protein loses its primary structure.

D At low acidity, the protein loses its secondary structure

4 The diagram below shows a molecular structure.



What bond is enclosed by the box in the diagram above?

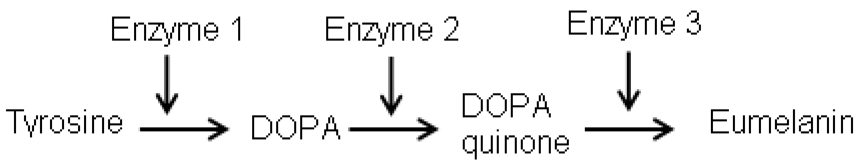
A Glycosidic bond

B Peptide bond

C Ester bond

D Phosphodiester bond

5 A part of the metabolic pathway to forming eumelanin in an individual is shown below.



The addition of substance **X** resulted in no change in the concentration of tyrosine, an accumulation of DOPA and a near absence of both DOPA quinone and eumelanin.

Further addition of DOPA quinone results in formation to eumelanin.

What does the information indicate about substance **X**?

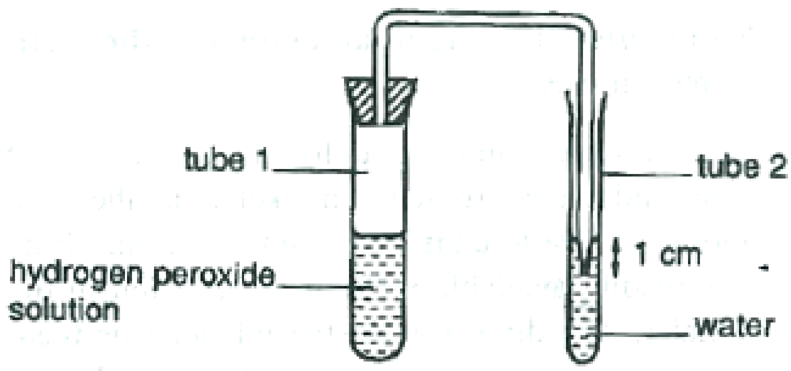
A It is an inhibitor of enzyme 1.

B It is an inhibitor of enzyme 2.

C It reacts with DOPA.

D It reacts with DOPA quinone.

6 Potato tubes contain an oxidoreductase enzyme that is able to catalyse the breakdown of hydrogen peroxide. The experiment is set up as shown below.



An investigation was carried to determine how differing concentration of hydrogen peroxide affects the rate of breakdown. 10 discs of 2 mm long potato slices were used each time.

The table below shows the results of the investigation.

Bubbles per minute	Concentration of hydrogen peroxide / mol dm^{-3}			
	0.5	1.0	1.5	2.0
Average number	4	9	16	20

How would the results differ (if any) if a single piece of 2 cm long potato was used instead?

A The average number of bubbles would remain the same.

B The average number of bubbles in all concentration would increase.

C The average number of bubbles in all concentration would decrease.

D The average number of bubbles would be the same in 0.5 mol dm^{-3} and 1.0 mol dm^{-3} hydrogen peroxide solutions.

7 The laws of inheritance derived by Gregor Mendel have become the backbone of classical genetics. Law of Segregation and Law of Independent Assortment make up the first two laws that Mendel posited.

Which of the following options show the correct stage of meiosis that explains the theory behind the proposed laws?

	Law of Segregation	Law of Independent Assortment
A	Anaphase I	Metaphase I
B	Anaphase I and II	Metaphase I
C	Anaphase II	Metaphase II
D	Anaphase I and II	Metaphase II

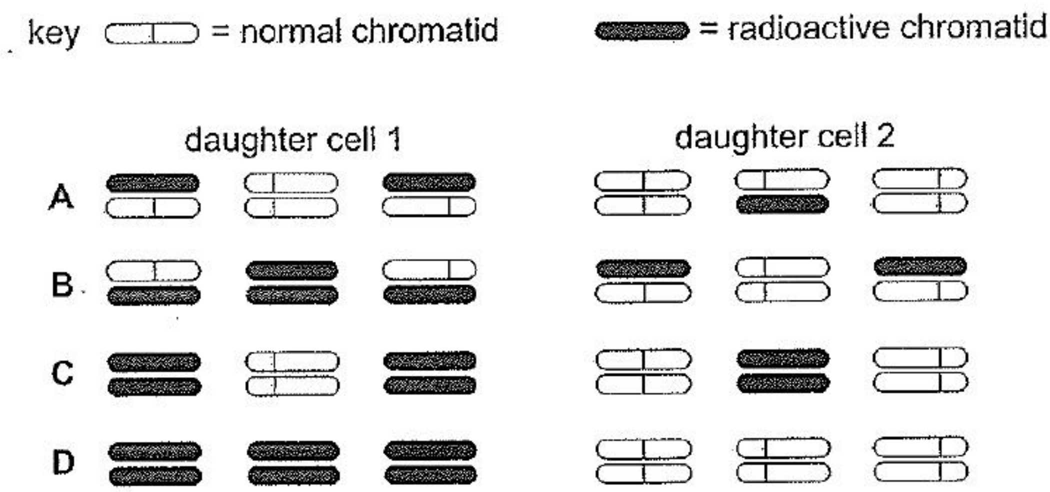
8 The following experiment was carried out.

1. Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive ^{15}N, so that all the DNA was labelled.

2. Cells in early interphase were then transferred to a medium with normal ^{14}N.

3. A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?



9 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

1 The anti-parallel nature of the DNA strands.

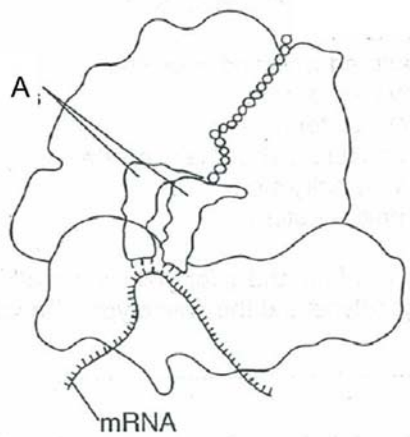
2 One strand is synthesized in 5' to 3' direction and another strand is synthesized in 3' to 5' direction.

3 DNA nucleotides are added to 3' end only.

4 One strand acts as template for leading strand while the other strand acts as template for lagging strand.

A 1 and 3
B 1 and 4
C 2 and 3
D 2 and 4

10 The diagram shows a process occurring at the rough endoplasmic reticulum.



Which of the following is true of molecule A?

A Its bases are complementary to those of the mRNA strand.

B It is synthesized by transcription of the genes in the nucleolus.

C It fits into another enzyme which has an active site for another molecule.

D There are less than 20 different types of them in the cytosol.

11 Ribonuclease is an enzyme that digests RNA.

The mRNA of the gene coding for a functional ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAUGAAACUGCU

A **non-functional** molecule of ribonuclease has been isolated. The first three amino acids of this non-functional molecule of ribonuclease are:

met – lys – leu –

The genetic code table is shown below.

first position	second position U	C	A	G	third position
U	phe phe leu leu	ser ser ser ser	tyr tyr STOP STOP	cys cys STOP trp	U C A G
C	leu leu leu leu	pro pro pro pro	his his gln gln	arg arg arg arg	U C A G
A	ile ile ile met	thr thr thr thr	asn asn lys lys	ser ser arg arg	U C A G
G	val val val val	ala ala ala ala	asp asp glu glu	gly gly gly gly	U C A G

Which event occurs to explain the information given above?

A A single base deletion mutation results in a frameshift mutation.

B A silent mutation occurs, but due to the degeneracy of genetic code, it results in different amino acids coded despite using the same mRNA codons.

C Translation started with ribosome binding to the second START codon.

D Two tRNAs have carried a different amino acid.

12 Two viruses **A** and **B**, one isolated from a prokaryote and another from eukaryote, have been identified.

It is found that treatment with reverse transcriptase inhibitors blocks the ability of **virus B** to infect cells but not **virus A**.

Which of the following conclusions can be drawn?

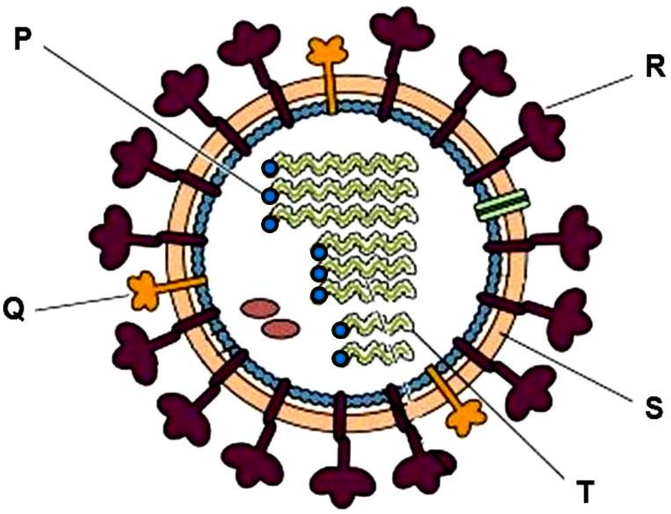
A Virus **A** has single-stranded RNA while virus **B** has double-stranded DNA.

B Virus **A** is a bacteriophage while virus **B** is a retrovirus.

C Virus **A** is a retrovirus while virus **B** is a bacteriophage.

D Virus **A** has double-stranded RNA while virus **B** has single-stranded DNA.

13 The diagram shows the structure of an influenza virus.



Which of the following statements concerning the lettered components are correct?

1 Mutations that disrupt the function of **P** will result in the inability of the virus to initiate infection in the host cell.

2 Influenza viruses are able to cause pandemics in humans due to changes in **Q** and **R** by a slight mutation.

3 **Q** and **R** are first synthesized, and embedded in the host cell surface membrane before budding takes place.

4 **S** is synthesized in the host cell using the host cell enzymes before assembly of the virus.

5 Different **T** from influenza strains infecting different animals can be assembled into one virus within one organism.

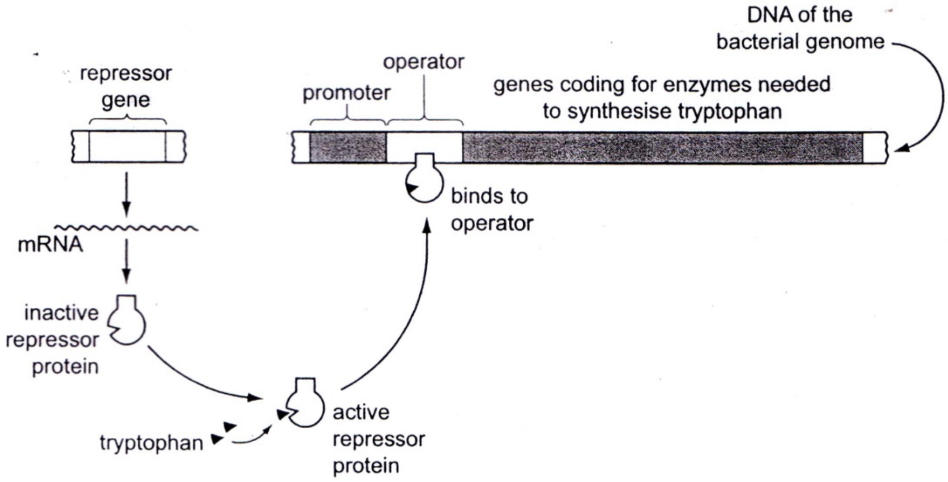
A 1, 3 and 5

B 1, 2 and 5

C 2, 3 and 4

D 3, 4 and 5

14 The diagram outlines the control of tryptophan production in *Escherichia coli*.



For a strain of *E. coli*, it was found that expression of *trp* operon was turned on in the presence of excess tryptophan. Which of the following is a likely explanation for this observation?

A A mutation occurred at the promoter.

B A repressor has bound to the promoter of the repressor gene.

C A mutation occurred at the gene coding for operator.

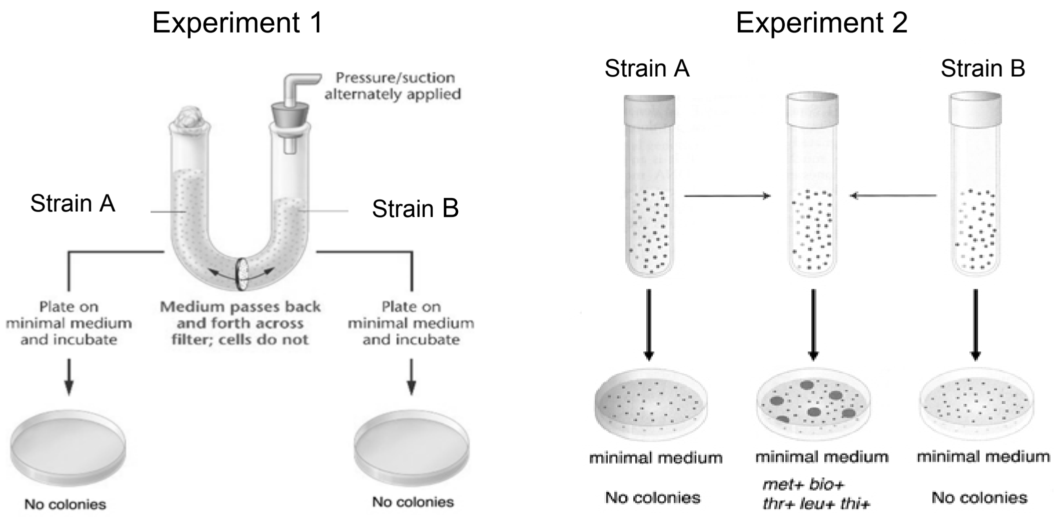
D Repressor protein binds permanently to tryptophan.

15 In the 1900s, two groups of researchers did experiments on how bacteria can undergo genetic recombination.

In both experiments, the researchers used the same 2 strains of bacteria. Strain A coded for 3 essential amino acids (thr^+ leu^+ thi^+) while Strain B coded for 2 essential amino acids (met^+ bio^+).

For the 2 strains of bacteria to grow on minimal medium, the bacteria need to encode for all 5 essential amino acids.

The two experimental set ups and results are shown in the diagram below.



Which of the following explains the results shown?

A The genes encoding for the amino acids are found on F plasmids.

B Mutation occurs at a higher rate when both strains of bacteria are mixed in the same culture.

C Specialised transduction was carried out by lambda phage to enable bacteria to survive in minimal medium.

D Transformation has occurred when the 2 strains of bacteria are mixed in the same culture

16 The electron micrograph below shows several labelled structures present in a mitochondrion.

Which of the statements below correctly describe the labelled structures?

1 The structure labelled **A** is the polypeptide chain.

2 The structures labelled **B** are polyribosomes which consist of many 70S ribosomes.

3 The structure labelled **C** is the 3' end of template DNA strand.

4 The structure labelled **C** is the 5' end of the mRNA strand.

A 1 and 2 only

B 1 and 4 only

C 2 and 3 only

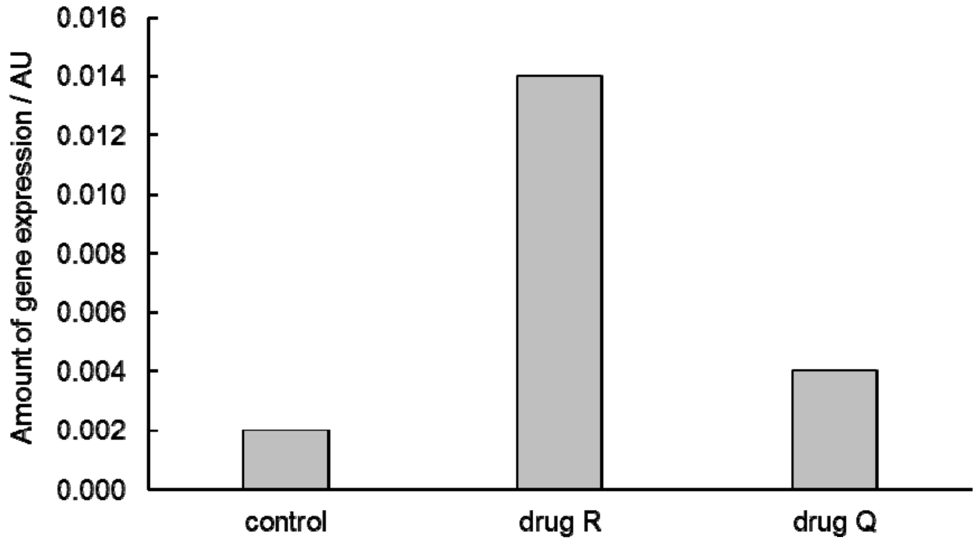
D 1, 2 and 4 only

17 Gene expression in eukaryotes can be regulated at the translational level.

Which combination of statements correctly describes eukaryotic translational control?

	Condition	**Effect**
A	Lack of translation initiation factor proteins	Inhibition of translation of selected mRNA
B	Presence of repressor proteins binding to 5'-UTR of selected mRNA	Prevents small ribosomal subunit from binding
C	Presence of repressor proteins to distal control elements	mRNA is translationally-repressed
D	mRNA with a long poly-A tail	mRNA is degraded slowly by restriction endonuclease

18 Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on expression of a gene. The graph shows the experimental results.



Which are possible explanations to the results shown?

1 Drug Q controls modification of histones, resulting in weaker binding of histones to DNA.
2 Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.
3 Drug R increases gene expression by preventing methylation at CpG islands at the promoter.
4 Inhibiting DNA methylation is more effective in increasing gene expression than inhibiting histone deacetylation

A 1 and 2 only
B 2 and 4 only
C 1, 2 and 3 only
D All of the above

19 A strain of mice has been genetically modified so that they display symptoms of a recessive sex-linked trait that causes death before the mice are able to breed.

In these mice, 33% of the alleles of this gene were recessive. In multiple crossings of carrier females and normal males, it was found that 50% of the males died, but no females.

What percentage of the alleles of this gene in the surviving offspring would be recessive?

A 0%
B 17%
C 20%
D 33%

20 In mice, the gene loci for mice coat pattern and whisker curvature are located on different chromosomes. For the gene locus controlling coat pattern, the allele for ‘dappled’ coat (**D**) is dominant to the allele for ‘plain’ coat (**d**). For the gene locus controlling whisker curvature, the allele for ‘straight’ whiskers (**W**) is dominant to the allele for ‘bent’ whiskers (**w**).

A male mouse with plain coat and bent whiskers was mated to a female dappled mouse with straight whiskers. The phenotypes and number of offspring mice having each phenotype was counted and recorded in the table below.

Phenotype	**Number of offspring**
dappled, straight whiskers female	52
dappled, straight whiskers male	39
plain, straight whiskers female	44
plain, straight whiskers male	48

In a separate cross, a female mouse with plain coat and bent whiskers was mated to a male dappled mouse with straight whiskers. The phenotypes and number of offspring mice having each phenotype was counted and recorded in the table below.

Phenotype	**Number of offspring**
dappled, straight whiskers female	24
plain, straight whiskers male	31

If the female parental mice from the first cross was mated to the male parental mice from the second cross and 96 offspring was produced, how many offspring mice would be expected to have dappled coat and straight whiskers?

A 24

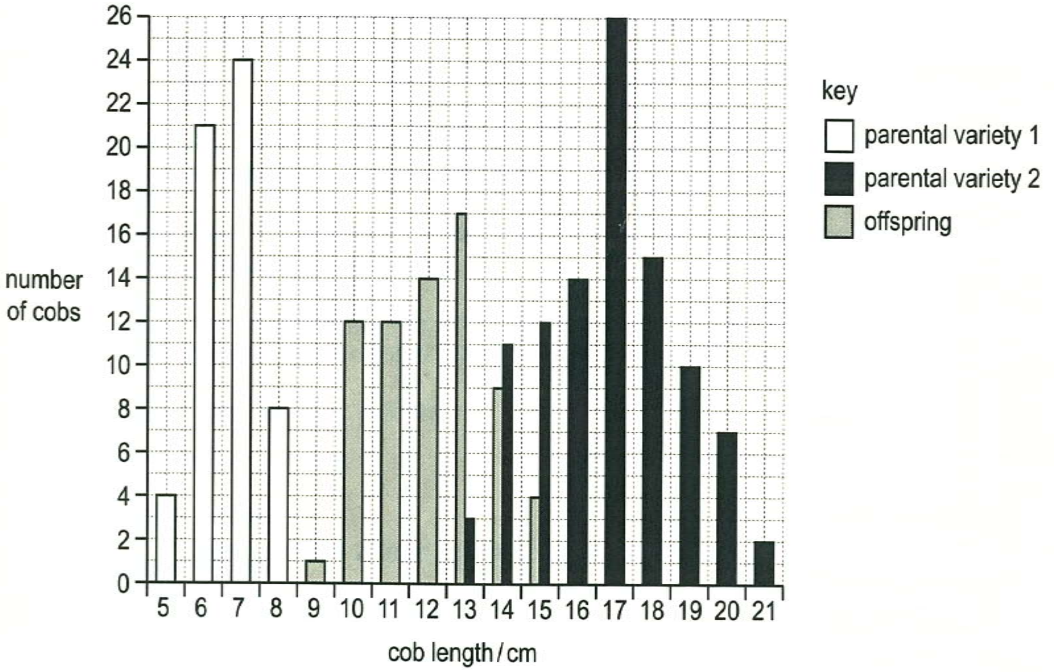
B 72

C 96

D None

21 Two pure-bred lines of different varieties of maize, which differed markedly in cob length, were crossed. The length of the cobs produced by the two parental varieties and their offspring were measured to the nearest centimetre. The number of cobs in each length category was counted.

The graph shows the results.



What is the cause of the phenotypic variation shown in cob length within each of the two parental varieties and their offspring?

A additive effect of different genes

B Linkage and crossing-over at meiosis

C Segregation and independent assortment of alleles

D Various environmental factors

22 A yellow flower, green-stemmed plant with the genotype **YYrr** was crossed with a white flower, red-stemmed plant with the genotype **yyRR**. The F_1 plants were allowed to self-fertilise. A $\mathbf{X^2}$ test was carried out on the results obtained for the F_2 generation. Part of the table of values for $\mathbf{X^2}$ is shown.

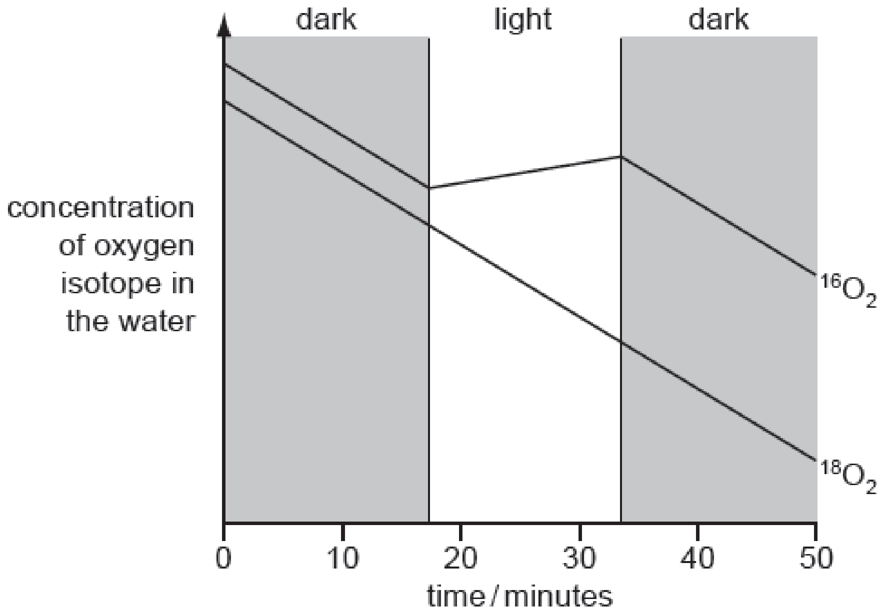
degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of $\mathbf{X^2}$ in this investigation was 10.6.

What is the probability of this value of $\mathbf{X^2}$ and do the results fit the expected ratio?

	Probability	**Results fit expected ratio**
A	Between 0.01 and 0.05	No
B	Between 0.01 and 0.05	Yes
C	Between 0.05 and 0.1	Yes
D	Between 0.1 and 0.5	No

23 The common isotope of oxygen is ^{16}O. ^{16}O and ^{18}O is bubbled through a suspension of algae for a limited period. After this, the concentration of these two isotopes of oxygen was monitored for the next 50 minutes whilst the algae were subjected to periods of light and dark. The results are shown in the diagram.



What is the best explanation for these results?

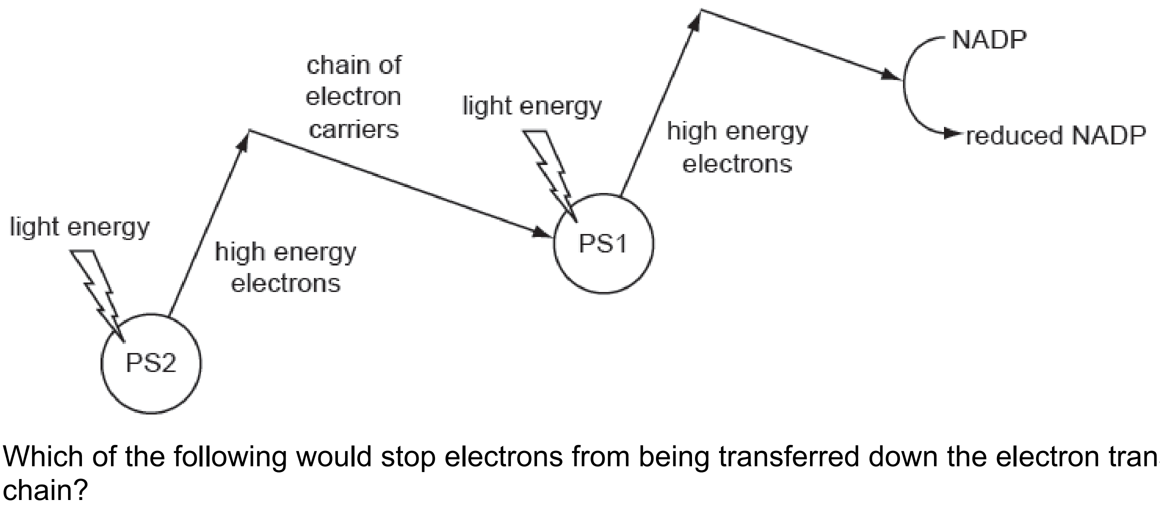
A $^{16}O_2$ was being absorbed at different rates in light and dark.

B The algae can distinguish chemically between the two isotopes.

C $^{16}O_2$ was being produced in photosynthesis but was not being absorbed in respiration.

D The algae produce oxygen from the water which is used in photosynthesis, but only in the light.

24 The diagram shows some of the processes in the light-dependent stage of photosynthesis.



Which of the following would stop electrons from being transferred down the electron transport chain?

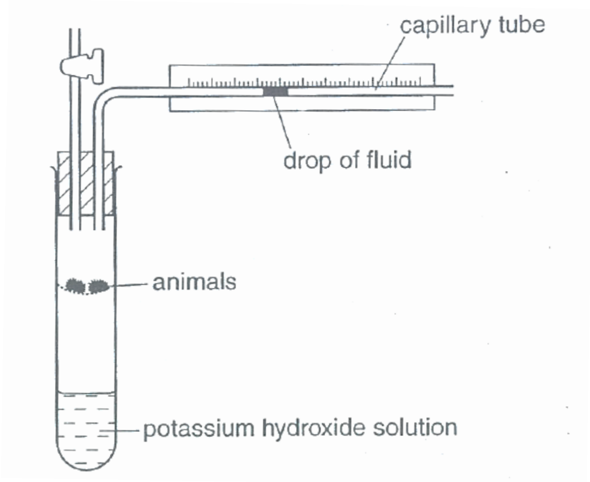
A Proton pump coupled to electron carriers were inhibited.

B Reduced NAD unable to transfer electrons to electron transport chain.

C Thylakoid membrane becomes 'leaky'.

D Bleaching of chlorophyll.

25 A respirometer was set up to investigate the rate of respiration as shown below.



At the end of the experiment, it was found that the drop of fluid moved to the left significantly more than expected. Mitochondria from the animals were isolated to investigate the cause.

Which of the following could be a possible observation made about the mitochondria which best explains the results?

A Cytochrome *b* in the electron transport chain was non-functional.

B Higher than normal amount of cristae observed.

C Proton pump is decoupled with the electron carriers.

D ATP synthase is hyperactive.

26 The enzyme phosphofructokinase is involved in the phosphorylation of hexose phosphate sugars during glycolysis. It is involved in the control of the rate of glycolysis, and thus respiration, by end-product inhibition.

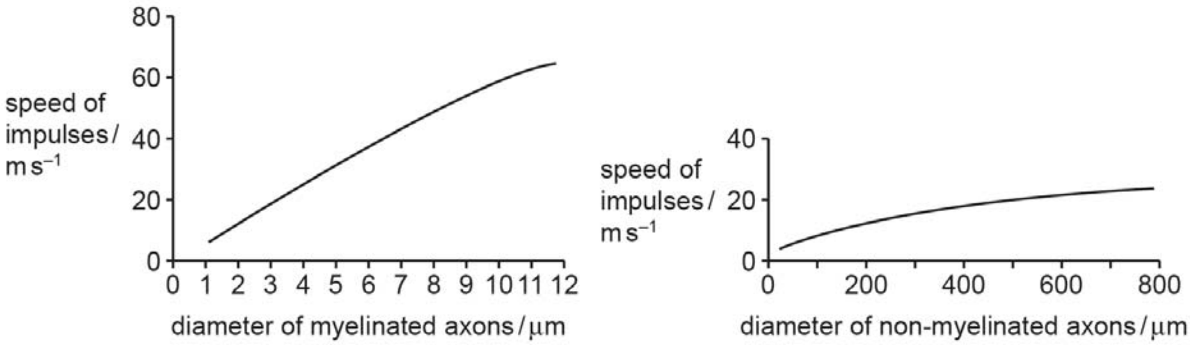
Deduce which of the following is a description of this enzyme.

	shape of binding site(s)	**substrate**	**products**
A	no allosteric site; active site is complementary to ATP and hexose	hexose	hexose phosphate
B	allosteric site is complementary to glucose; active site is complementary to hexose phosphate	hexose phosphate	hexose phosphate
C	allosteric site is complementary to ATP; active site is complementary to ATP and hexose phosphate	hexose phosphate	hexose bisphosphate
D	no allosteric site; active site is complementary to hexose bisphosphate	hexose bisphosphate	Two triose phosphate

27 The ability of organisms to respond rapidly to stimuli is limited by the speed of the impulses in their neurons.

The axons of invertebrate neurones lack a myelin sheath. The axons of most vertebrate neurones are myelinated.

The graphs show the speed of impulses in these two types of axon.



Which statements about these data are correct?

1 The action potential in myelinated axons is greater than the action potential in non-myelinated axons.
2 The speed of impulses is changed by the diameter of the axon.
3 Increasing the diameter of a myelinated axon causes a greater change to the conduction speed than increasing the diameter of a non-myelinated axon.
4 The presence of myelin increases the speed at which impulses are conducted.

A 1, 2 and 3 only

B 1, 2 and 4 only

C 2, 3 and 4 only

D 3 and 4 only

28 Gamma-aminobutyric acid (GABA) is a neurotransmitter that functions as a signalling molecule in the central nervous system. GABA binds to a receptor protein located in the plasma membrane of target cells as shown in Fig.1. Binding of a GABA molecule opens a channel that allows chloride ions (Cl^-) to enter the cell.

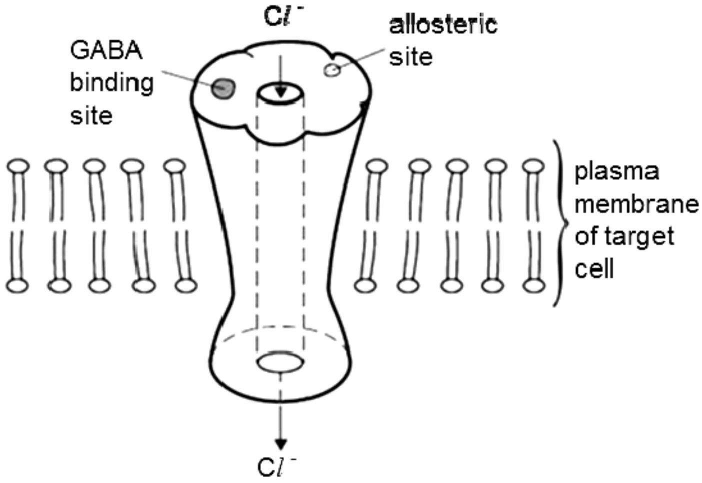


Fig. 1

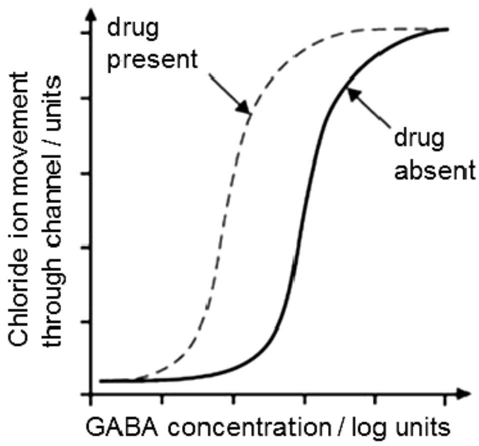


Fig. 2

In a treatment for seizures, Valium, a drug, acts as allosteric regulator by binding at a site that is away from the GABA-binding site. Fig. 2 shows the movement of chloride ions through the channel as GABA is increased with and without the drug being present.
Which statements are correct?

1 The GABA receptor is a multimeric protein.

2 Binding of Valium enhances the opening of the channel.

3 A depolarization occurs in the target cell.

4 Binding of Valium helps to open the channel even if GABA is not present.

A 1 and 2 only

B 2 and 3 only

C 1, 2 and 4 only

D 2, 3 and 4 only

29 The mean levels of glucose and insulin in two groups of people are sampled and the results are shown in the table below.

	group P			**group Q**		
time after ingestion of 75g of glucose (min)	0	15	60	0	15	60
plasma glucose (mmol / L)	5.3	7.9	7.8	5.3	7.7	6.4
plasma insulin (pmol / L)	165	208	210	171	215	180

Which of the following statements about the above information is/are false?

1 People from group P lack beta cells.

2 Increased gluconeogenesis and glycogenolysis occur in people from group P after ingestion of 75g of glucose.

3 More glucose can enter the target cells via facilitated diffusion in people from group Q due to the increase in numbers of the glucose transporters upon binding of insulin.

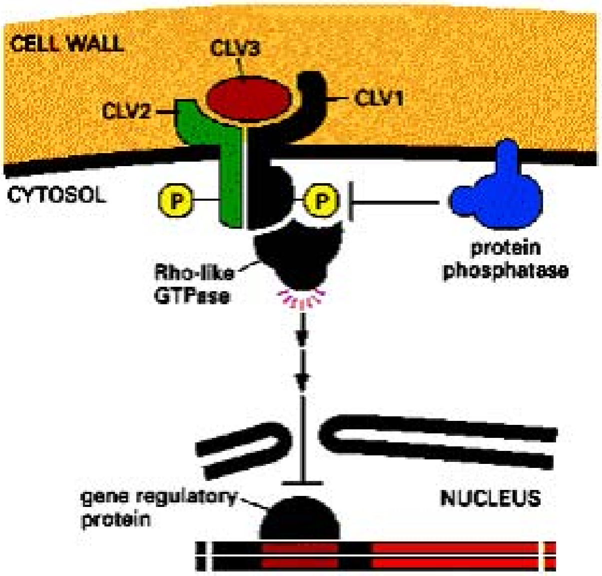
A 3 only

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

30 The diagram below shows the cell signaling pathway involved in cell proliferation and differentiation in the shoot meristem.



Legend:
⊣ inhibition

Which one of the following sequences regarding the activation of the above signaling pathway is **correct**?

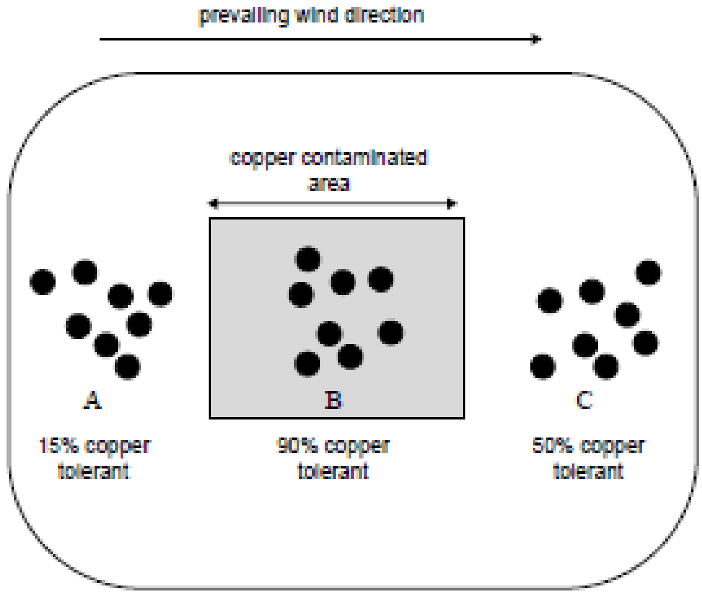
A binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → inhibition of gene transcription

B dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription

C binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription.

D dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → inhibition of gene transcription

31 The diagram below shows an aerial view of a region which included a section of copper contaminated soil (shaded grey on the diagram). The black circles represent populations of grass growing in areas A, B and C. The diagram also includes the percentage of copper resistant plants in areas A, B and C. Tolerance to copper in the soil is genetically determined.



What is a reasonable conclusion from the information?

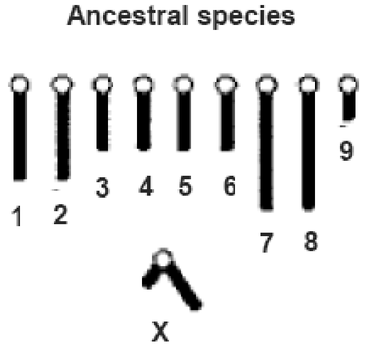
A High levels of copper tolerance would be a selective advantage for all plants in areas A, B and C.

B All plants in the copper contaminated area would be homozygous at the gene locus for copper tolerance.

C Gene flow is occurring between populations in areas B and C.

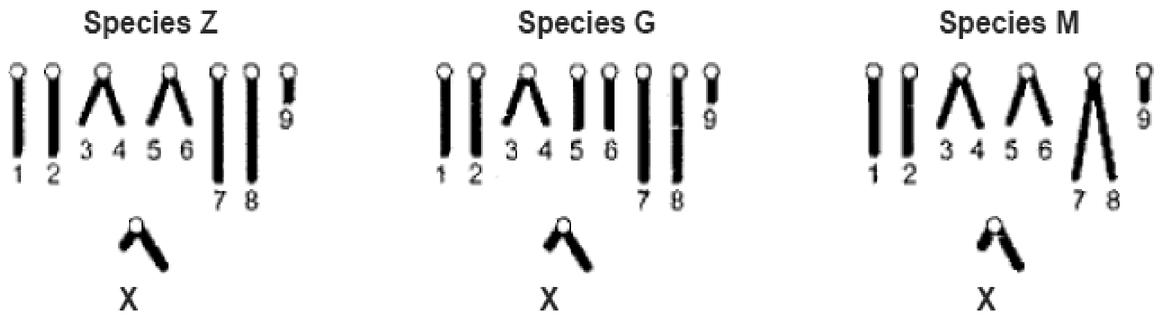
D The difference in copper tolerance between populations in areas A and C is the result of mutation.

32 In some Australian insects, new species have arisen through changes that occurred to chromosomes in an ancestral species. Such changes may involve the joining together of chromosomes, the loss of whole or parts of chromosomes, and rearrangement of the genetic material within chromosomes. One ancestral species has the following haploid set of chromosomes.



As the changes in chromosomes accumulate, a number of different species can result from a single ancestral species.

Three species that have evolved from the ancestral species shown above have the haploid sets of chromosomes shown below.



The most likely order of evolution of these species is

A ancestral species, species Z, species G, species M.

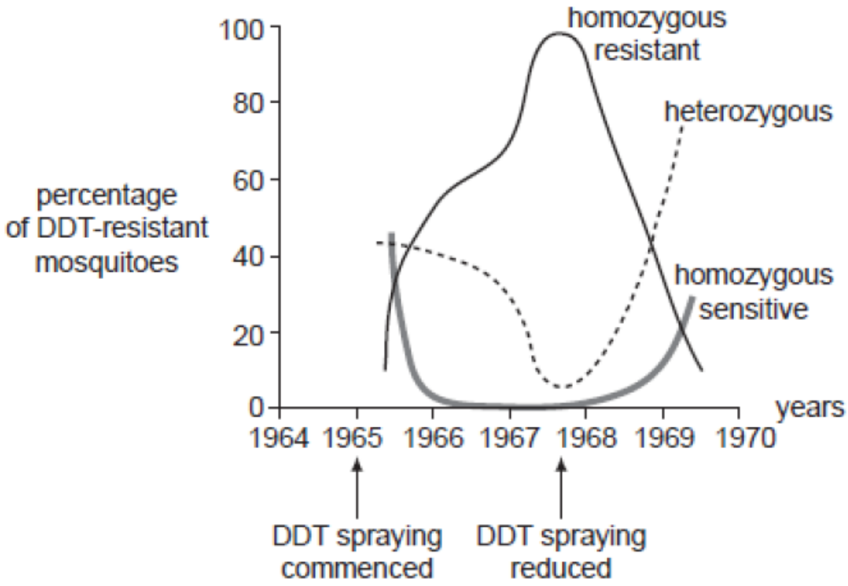
B ancestral species, species G, species M, species Z.

C ancestral species, species M, species G, species Z.

D ancestral species, species G, species Z, species M.

33 In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles. Insects with DDT resistance also has reduced metabolism.

Over several years genotypic frequencies were measured in a population of mosquito larvae. The graph below shows the results.



Analysis of the graph reveals that in the population

A the homozygous resistant insects compete better for resources than heterozygous ones, until heterozygous advantage occurs when DDT spraying is reduced.
B there were no alleles for sensitivity present in the population in 1967, but they were created by gene mutation after 1967.
C the number of alleles for resistance was equal to the number for sensitivity in 1966.
D the homozygous resistant genotype was unable to produce offspring at low spraying levels.

34 The rate of divergence of different proteins is shown in the table below.

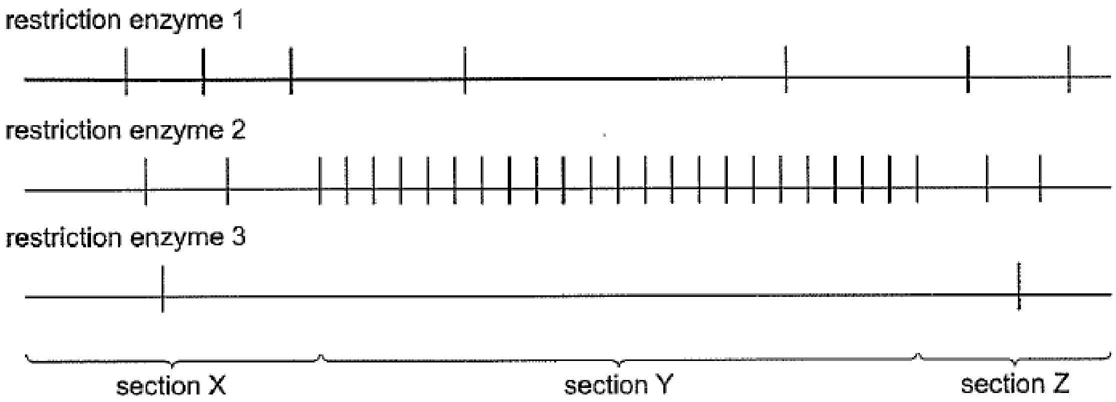
Protein	**Amino acid substitutions per site per billion years**
Fibrinopeptides	8.3
Lysozyme	2.0
α-globin	1.2
Insulin	0.44
Cytochrome *c*	0.3
Histone H4	0.01

What does this data tell you about the neutral theory of molecular evolution?

A In fibrinopeptides, there are more mutations in the coding regions than non-coding regions as the protein has the largest amino acid substitutions per site per billion years.
B More essential proteins allow for fewer amino acid substitutions per site than others.
C The rate of neutral mutations that result in the amino acid substitutions for all proteins is the same.
D All of these proteins are found in a common ancestor of all Earth's known living organisms before they started evolving at the molecular level.

35 A linear fragment of DNA made up of three sections, X, Y and Z, was cut using three different restriction enzymes.

The vertical lines on the diagrams show the sites at which the different restriction enzymes cut this DNA fragment.



Which conclusion is correct?

A Isolation of section Y requires the use of another restriction enzyme.

B Sections Y and Z are composed of many repeating sequences of DNA.

C Digestion with restriction enzymes 1 and 3 together will produce nine separate DNA fragments.

D Restriction enzymes 1, 2 and 3 bind to DNA via complementary base pairing.

36 Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

The dashed lines in the template sequence stand for a long sequence of bases that is to be amplified.

Template

5' ATTCGGACTTG - - - - - - - - - GTCCAGCTAGAGG 3'
3' TAAGCCTGAAC - - - - - - - - - CAGGTCGATCTCC 5'

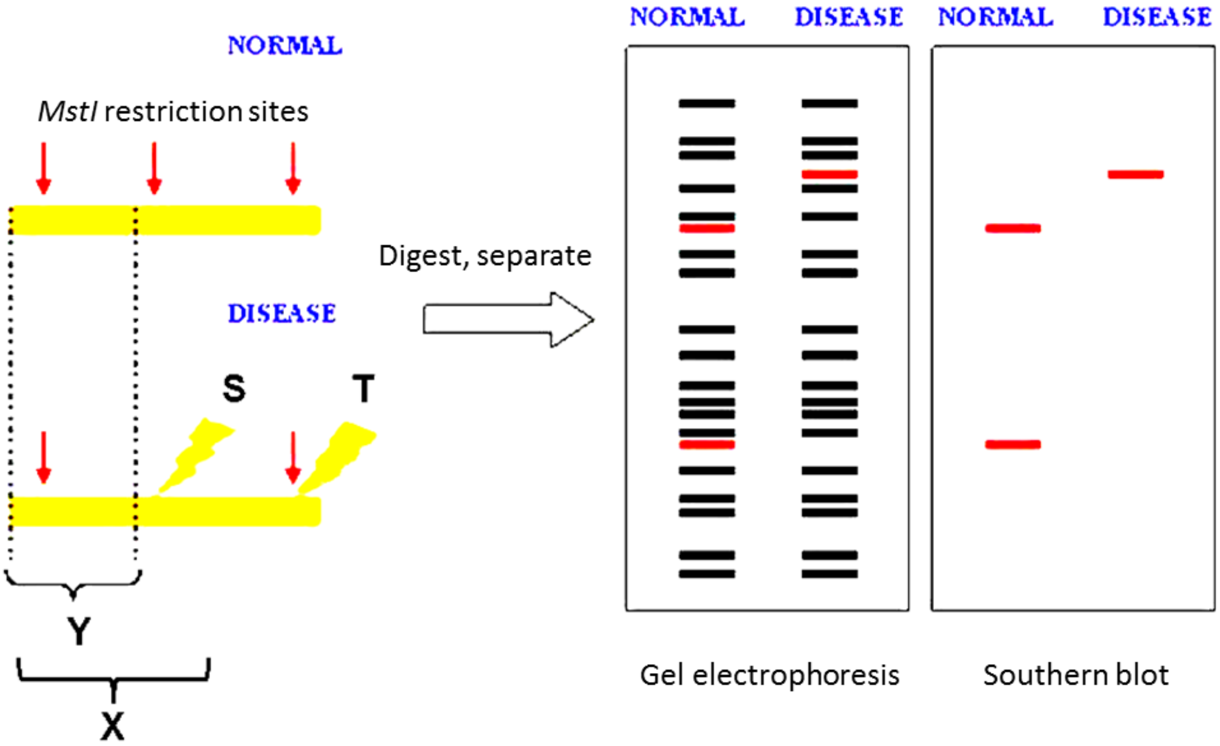
A 5’ GTCCAGC 3’ & 5’ CCTGAAC 3’

B 5’ ATTCGGA 3’ & 3’ GATCTCC 5’

C 5’ GGACTTG 3’ & 5’ GCTGGAC 3’

D 5’ AUUCGGA 3’ & 3’ GAUCUCC 5’

37 A difference in homologous DNA sequences can be detected by the presence of fragments of different lengths after digestion of DNA samples with specific restriction endonucleases and using a labelled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis, thus revealing a unique blotting pattern, as shown in the figure below.



Which combination of probe and mutation correctly explains the band patterns in the Southern blot after treatment with *MstI* restriction enzyme?

	Probe used for hybridization to digested fragment	Location of point mutation in DNA sequence in the disease
A	X	T
B	Y	T
C	X	S
D	Y	S

38 Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell.

Which statement explains this?

A Different stem cells have only the genes required for their particular cell line.

B Genes not required for a particular cell line are methylated.

C Genes not required for a particular cell line are removed using restriction enzymes.

D mRNA that is not required for a particular cell line is destroyed.

39 Increased genetic diversity following extended time in a plant tissue culture is a problem called

A gene alteration

B culture shock

C acclimatisation shock

D somaclonal (somatic) variation

40 One type of genetically modified corn has

- a gene for the production of Bt toxin which protects the plant against a specific insect;
- a 'pat'-gene for tolerance to the herbicide 'Basta'. This gene is used to select plants with the Bt toxin gene;
- an 'amp'- gene which was introduced in the plant together with the Bt toxin gene. This gene gives resistance to the antibiotic ampicillin.

There is concern that the 'amp' gene may transfer to enterobacteria in the human intestine during nucleic acid digestion making treatment with ampicillin ineffective for diseases caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

A An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.

B Bacteria cannot take up any DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.

C 50% of the enterobacteria isolated from humans are 'amp' resistant.

D All plant DNA is digested and destroyed in the human intestine during digestion of plant cells by enzymes including human nuclease enzymes.